Lead-Lag Compensator

We saw that the lead compensator was used to improve the relative stability, that is, the location of the closed loop dominant poles. In this case, the gain was adjusted to satisfy the magnitude condition of the root locus and not the steady state error. We assume, then, that the gain obtained by satisfying the magnitude condition is good enough for the steady state error. (For example, if the plant is type 1, the closed loop system has zero error to the input step regardless of the gain. The error is always zero because of the integrator in the plant).

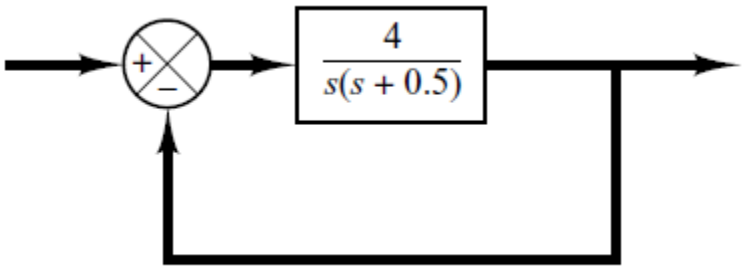
The lag compensator was used to improve the steady state error when the location of the closed loop dominant poles was good. The idea of the lag compensator is to introduce a compensator that did not affect much the root locus around the desired poles but will allow us to increase the gain to improve the steady state error.

What would we do, then, when we have a system for which we need to improve both the relative stability (pole location) and the steady state error? Simple answer, we use both compensators as follows:

1. First, design the lead compensator to satisfy the required closed loop pole location as done before to satisfy the required location for the closed loop dominant poles. This controller disregards the steady state error.
2. With the lead controller applied to the plant, find the gain that the lag controller must provide to satisfy the steady state error specification.
3. Design the lag controller as done before. Make sure the contribution (magnitude and angle) of the lag controller is negligible so it won't affect the location of the closed loop desired poles that the lead controller fixed.
4. You are done.

Let's do the following example:

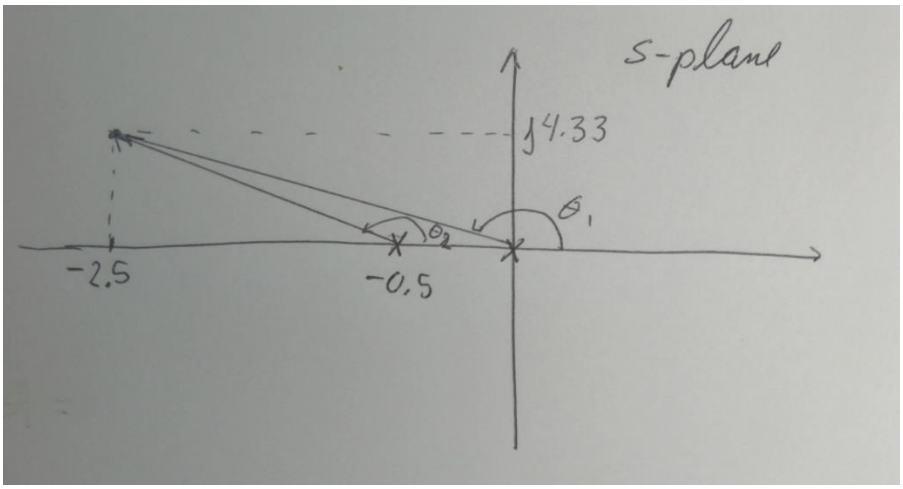
Consider the control system shown in the following figure:



Design a controller such that the closed loop poles have a damping ratio equal to 0.5 and an undamped natural frequency of 5 rad/s. Also, in order to have a small error to a ramp, it's required that the velocity constant be 80 s^{-1} .

To solve this problem, it's convenient to determine if the desired closed poles belong to the root locus without compensation. In the s-plane, we draw our pole-zero diagram of the open loop original system knowing that the desired poles are:

$$-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -2.5 \pm 4.33j$$



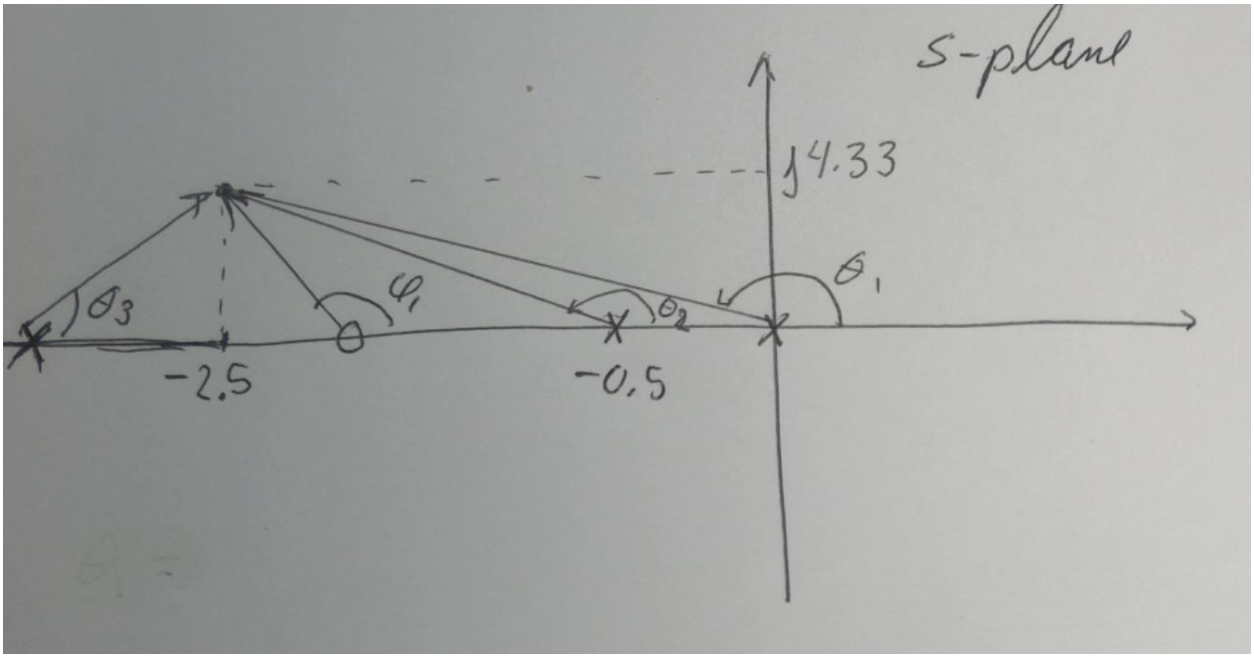
A little of trigonometry tells us that, $\theta_1 = 120°$ and $\theta_2 = 114.792°$, therefore

$$-\theta_1 - \theta_2 = -234.792°$$

which tells us that, that the desired closed loop poles are not actually in the root locus because the angle condition is not satisfied. For the angle condition to be satisfied, the controller has to supply an angular contribution of +54.792 because -234.792 + 54.792 = -180. The angle +54.792 is the angle deficiency.

Between the phase lead and the phase lag controllers, only the phase lead controller adds positive angular contribution. Following the steps of the lead-lag compensator, we are going to design first the lead compensator. Placing the pole and the zero of the lead compensator in our

open loop pole-zero diagram, we obain:



where, $\varphi_1 - \theta_3 = 54.792°$. We saw in class two methods to design the lead controller. The first method guarantees (so I told you) the maximum gain but is more complicated. In the second method you fix the zero location, and then find the pole; this method is more straightforward. For the sake of illustrating the lead-lag design process, we are going to ease our lives by choosing the second method.

Pick the zero of the lead compensator at s =-0.5 (you could have chosen many other zeros). This zero will cancel out the pole of the plant at the same location. Once this zero is chosen, we can find the pole such that the total angle contribution of the lead compensator is $54.792°$:

$$\varphi_1 = \theta_2 = 114.792°$$

because the compensator zero coincides with the plant pole at s=-0.5. Therefore, the compensator pole angular contribution is:

$$\theta_3 = 114.792° - 54.792° = 60°$$

and the compensator pole distance from the origin is

$$p = 2.5 + \frac{4.33}{\tan 60°} = 5$$

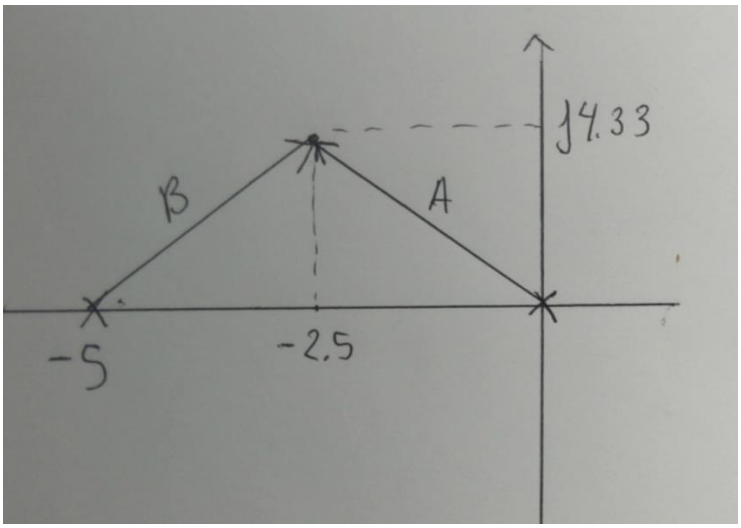
Hence the lead compensator is

$$G_A(s) = k_A \frac{(s+0.5)}{(s+5)}$$

We find the gain k_A by using the magnitude condition as said previously. Very likely, this gain won't satisfy the velocity constant requirements, but we need to verify that to make sure. If so, we don't have to calculate the lag compensator.

$$\left| k_A \frac{(s+0.5)}{(s+5)} \frac{4}{(s+0.5)s} \right|_{s=-2.5+j4.33} = 1$$

We can do that also with the help of some graphics (notice that a zero and a pole has been cancelled out)



$$\left| \frac{4k_A}{AB} \right| \Rightarrow k_A = \frac{AB}{4} = 6.25$$

Considering only the lead compensator, the velocity constant is

$$K_v = \lim_{s \to 0} s \left[\frac{6.25(s+0.5)}{(s+5)} \right] \left[\frac{4}{s(s+0.5)} \right] = 5$$

The velocity constant is then too low. We need a factor of 16 to raise it to the desired valued of 80. This extra gain of 16 can be provided by the lag compensator without affecting too much the pole location achieved with the lead compensator.

$$G_R(s) = K\frac{(Ts+1)}{(\beta Ts+1)} = \frac{K}{\beta}\frac{\left(s+\frac{1}{T}\right)}{\left(s+\frac{1}{\beta T}\right)}$$

The gain K is calculated to satisfy the steady state error required (in this case, in terms of $K_v = 80$). As determined before, K must be equal to 16. We are going to select $\beta = K = 16$; therefore,

$$G_R(s) = \frac{\left(s+\frac{1}{T}\right)}{\left(s+\frac{1}{\beta T}\right)}$$

In order not to affect the pole location achieved with the lead controller, we need to guarantee that $G_R(s_1) \approx 1$ where s_1 is one of the desired complex conjugate poles:

$$\left.\frac{\left(s+\frac{1}{T}\right)}{\left(s+\frac{1}{\beta T}\right)}\right|_{s_1=-2.5+j4.33} \approx 1$$

that is, $|s_1| \gg 1\backslash T => \quad 5 \gg 1/T$. We can choose $T = 5$, for example. Then, $\beta T = 16 * 5 = 80$:

$$G_R(s) = \frac{s+0.2}{s+0.0125}$$

Using Matlab or any other technique at your disposal, you can find:

$$\left|\frac{s_1+0.2}{s_1+0.0125}\right| = 0.9818$$
$$\angle\left(\frac{s_1+0.2}{s_1+0.0125}\right) = -1.9°$$

The angle value is small (between 0 and -5), and the magnitude is very close to 1. We do not expect the lag compensator to affect the root locus around $s_1 = -2.5 + j4.33$ so much.

Finally, our lead-lag compensator is:

$$G_c(s) = 6.25\left(\frac{s+0.5}{s+5}\right)\left(\frac{s+0.2}{s+0.0125}\right)$$

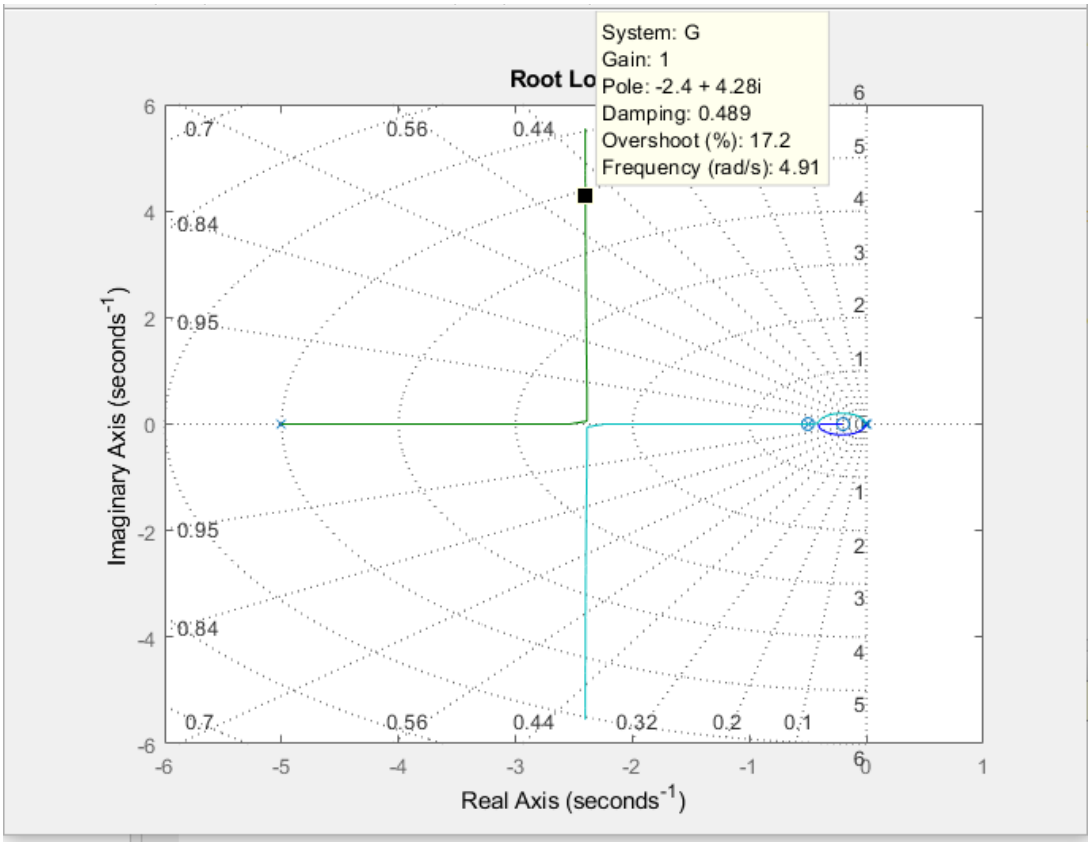
We also can verify the final velocity gain:

$$K_v = \lim_{s\to 0} s\left[\frac{6.25(s+0.5)}{(s+5)}\right]\left[\frac{s+0.2}{s+0.0125}\right]\left[\frac{4}{s(s+0.5)}\right] = 80$$

as desired. The error to a ramp will be 1/80=1.25%

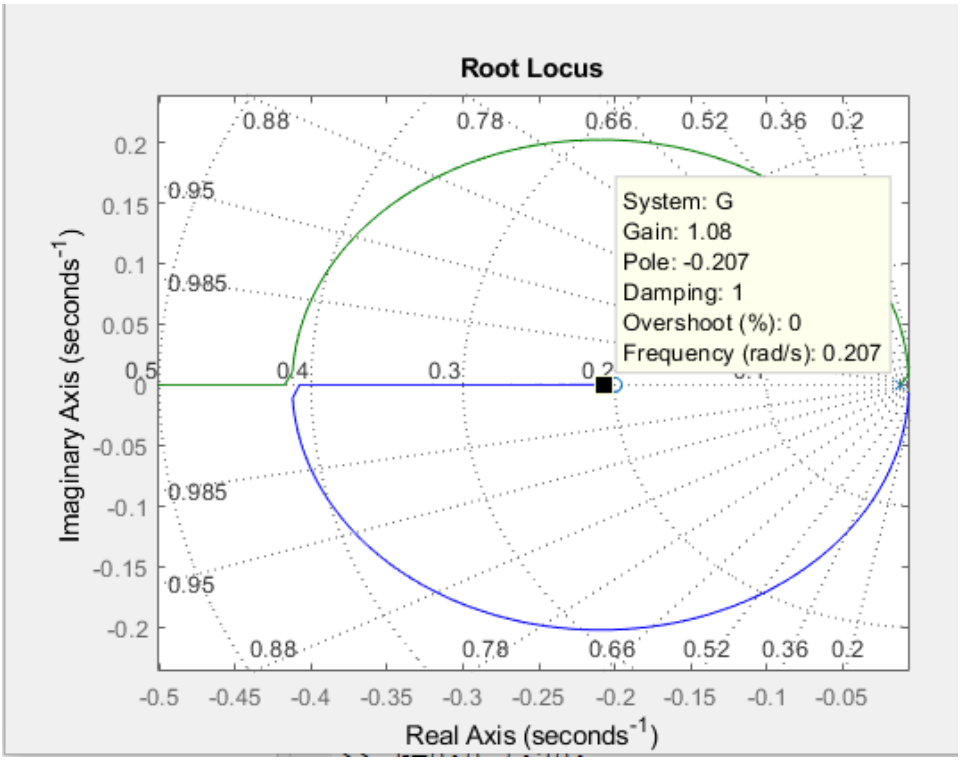
A Note about cancelling poles: In this problem we picked a compensator zero to cancel a plant pole. We need to be careful with this technique. Never try to cancel a pole that is too close to the imaginary axis or on the imaginary axis. And never, never, never, unless you want to wake the wrath of the universe, cancel a pole on the right half plane (unstable pole).

The root locus of the compensated system is found through Matlab. The above transfer function was entered; this means that the final pole location can be seen when the root locus gain in Matlab equals one:



You can see how the pole location changed with the introduction of the lag compensator. Notice also that on the right of the final poles, there is more root locus. This means that there are closer poles to the jw axis. These closer poles are the dominant poles. Can we trust our design to determine the real behavior of the system if they are not the dominant poles? In

general, the answer is no, but we need to take a look at that small region close to the origin. When we zoom out that region, this is what we see:



Approximately, at the gain of one, there is the most dominant pole at -0.207. This pole should dictate the behavior of the closed loop system, but it doesn't, why? Because the closed loop transfer function has a zero at -0.2 that practically cancels out this pole, so its effect is negligible. Remember, the zeros of the open loop transfer function are also zeros of the closed loop transfer function.